Counting Ehrhart functions of half-integral polygons

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Abstract

Fix an integer $i \geq 2$. We count the distinct Ehrhart functions of convex half-integral polygons with exactly i interior lattice points and at least two boundary lattice points. The number is $(9i^3 + 72i^2 + 175i + 106)/2$, as proposed by Bohnert and Springer. We describe the complete set of functions by their area and their boundary counts before and after doubling. Convex section estimates isolate two exceptional families, and explicit polygons realize every remaining signature. The proof uses the published complete computational classification for $2 \leq i \leq 16$, five published minimum-width bounds, and a finite list of exact rational inequality certificates. Integral polygons and period collapse are included, and no restriction is imposed on the arrangement of the interior lattice points.

1 Introduction

For a rational polygon P, its Ehrhart function records the lattice points in every integer dilate:

$$L_P(t) = \#(tP \cap \mathbb{Z}^2), \qquad t \in \mathbb{Z}_{\geq 0}.$$

For lattice polygons, Pick's formula determines this polynomial from the interior and boundary counts. Scott's inequality bounds the possible boundary counts when the interior count is fixed [1]. Thus counting the functions is much simpler than classifying the polygons themselves.

For half-integral polygons, meaning polygons whose vertices lie in $\frac{1}{2}\mathbb{Z}^2$, there are two polynomial constituents. The boundary count of the doubled polygon supplies an additional parameter. Bohnert's coefficient formula makes these parameters explicit [2, Proposition 2.1]. Bohnert and Springer established sharp boundary and area bounds [3, Theorems 1.1 and 1.4]; these bounds constrain the coefficients but do not determine which intermediate values of the doubled boundary count occur. Their published classification enumerates all functions through $i = 16$ and proposes a cubic formula for the number with at least two boundary lattice points [4, Conjecture 6.8].

We prove that formula.

Theorem 1.1. *For every integer $i \geq 2$, the number of distinct functions L_P among all two-dimensional convex polygons P with vertices in $\frac{1}{2}\mathbb{Z}^2$, exactly i interior $\mathbb{Z}^2$ points and at least two boundary $\mathbb{Z}^2$ points is*

$$\mathcal{E}(i) = \frac{9i^3 + 72i^2 + 175i + 106}{2}.$$

Here half-integral includes integral polygons. Distinctness means equality of the entire function on $\mathbb{Z}_{\geq 0}$, so a function with period collapse is counted only once. The class is exactly that in the published conjecture; in particular, the interior points need not be collinear.

There are two geometric obstructions to a direct count from the marginal area bounds. First, a polygon can have an integer hull which is a segment. A thin family then has a smaller doubling

interior count and must be treated separately. Second, the upper bound coupling area and doubled boundary count has a width-two equality family. We identify both families. Outside them, section estimates give a finite set of inequalities. Uniform constructions then show that those inequalities describe all signatures, rather than only necessary restrictions.

The proof retains explicit computational inputs. Small interior counts use the complete published classification, and a near-area estimate uses five minimum-width maxima from the same maximal-polygon catalogue. The wide section inequality is proved by rational linear combinations for finitely many width indices and by an analytic bound for the remaining indices. The all-parameter constructions and the count itself are proved symbolically. These roles are distinguished in Section 6.

We first give the signature classification, then prove its necessary bounds and realize every signature. The final count is in Section 7. Throughout, area is Euclidean area, lattice points without a qualifier belong to $\mathbb{Z}^2$, and all polygons are convex and two-dimensional.

2 Signatures and the classification

Write

$$i = i(P), \quad b = b(P), \quad Q = 2P, \quad A = 8\,\mathrm{area}(P) = 2\,\mathrm{area}(Q), \quad B = b(Q).$$

Thus A, B are integers. Bohnert's coefficient formula [2, Proposition 2.1] gives

$$L_P(t) = \begin{cases} \dfrac{A}{8}t^2 + \dfrac{b}{2}t + i + \dfrac{b}{2} - \dfrac{A}{8}, & t \text{ odd}, \\ \dfrac{A}{8}t^2 + \dfrac{B}{4}t + 1, & t \text{ even}. \end{cases} \tag{1}$$

The even formula includes $t = 0$. At fixed i, different triples (A, b, B) give different functions: the even constituent determines A, B and the odd linear coefficient determines b. Period collapse occurs precisely when

$$B = 2b, \qquad A = 8i + 4b - 8. \tag{2}$$

Define three sets of integer triples. For the *body*, let $2 \le b \le 3i + 6$, put

$$q(b) = \begin{cases} 8, & b \le 3i + 4, \\ 7, & b = 3i + 5, \\ 6, & b = 3i + 6, \end{cases}$$

and require

$$6i + 4b - 6 \le A \le 12i + 2b + q(b), \tag{3}$$

$$A \equiv B \pmod 2, \tag{4}$$

$$2b + (A \bmod 2) + 2\delta \le B \le \min\{A - 6i - 2b + 6,\ 3i + b + 6,\ 12i + 4b + 10 - A\}. \tag{5}$$

Here $\delta = 1$ if $A = 12i + 2b + 8$ and $b \ne 2i + 4$, or if $b = 2, A = 12i + 10$; otherwise $\delta = 0$. The *top* is

$$(A, b, B) = (12i + 2b + 8 - d,\ b,\ 2b + 4 + d), \quad d \in \{0, 1, 2\}, \quad b \ge 2, \quad b + 2d \le 2i + 2. \tag{6}$$

The *line* is

$$(A, b, B) = (A, 2, A - 4i), \qquad 4i + 4 \le A \le 8i + 8. \tag{7}$$

Let C_i be their union.

Theorem 2.1. *For every $i \geq 2$, the signatures (A, b, B) of the polygons in Theorem 1.1 are exactly C_i.*

We prove necessity for $i \geq 17$ in the next two sections. For $2 \leq i \leq 16$ we use the complete published classification as specified in Section 6. Section 5 proves attainment for all $i \geq 2$.

We use the following published marginal bounds. For a nonintegral half-integral polygon with $i \geq 2$,

$$b \leq 3i + 6, \qquad A \leq 12i + 2b + q(b) \tag{8}$$

by [3, Theorems 1.1 and 1.4]. At $b \geq 3$, its Proposition 3.3 also gives

$$B \geq 2b + 2 \quad \text{if } A = 12i + 2b + 8,\ b \neq 2i + 4. \tag{9}$$

Only this necessary assertion is used from that proposition. For integral polygons, Pick and Scott instead give $A = 8i + 4b - 8,\ B = 2b,\ b \leq 2i + 6$; these satisfy the marginal bounds and cannot reach the exceptional maximum area in (9). In general $B \geq 2b$: between successive original boundary points is a nonoriginal half-grid boundary point. Pick on Q gives

$$A = 2i(Q) + B - 2, \qquad L_P(2) = \frac{A + B + 2}{2}, \tag{10}$$

and hence (4).

3 Doubling and convex sections

3.1 The lower doubling bound

Lemma 3.1. *Either $i(Q) \geq 3i + b - 2$, or the signature belongs to (7).*

Proof. Let $S = P \cap \mathbb{Z}^2$, $K = \operatorname{conv}(S)$. If K is two-dimensional, triangulate it with every point of S as a vertex. All edges are primitive. Writing $h = i(K)$, the number of edges is

$$E = 2i + 2b + h - 3.$$

This follows from the fan triangulation formula $E = 3\#S - b(K) - 3$, since $b(K) = i + b - h$; successive insertions preserve it.

Discard boundary edges whose two endpoints are boundary points of P. All other edge interiors lie in $\operatorname{int}(P)$. Their primitive midpoints are nonintegral and distinct, so they yield distinct interior points of Q, in addition to the i doubled original interior points. If s edges were discarded, $i(Q) \geq i + E - s$. For $h \geq 1$, the cyclic boundary has at most b discarded edges. For $h = 0$, it contains both original interior and boundary types, so a boundary-type run of length r contributes only $r - 1$ discarded edges; thus $s \leq b - 1$. Both give the required bound.

If K is a segment, an integer affine unimodular transformation puts its line at $y = 0$. Since P has interior points on that line, it extends on both sides. All its original boundary points are the two intersection endpoints. Thus $b = 2$ and we may take them as $(0, 0), (R, 0)$, $R = i + 1$. A vertex of height $H \geq 3/2$ would give at $y = 1$ a triangle section of length $R(1 - 1/H) \geq 1$, containing an off-line original point. The reflected argument excludes heights at most $-3/2$.

If a vertex has height 1, its horizontal coordinate x is a nonintegral half-integer. Its doubled triangle contains R interior fine points on $y = 1$, while the middle row contains $2R - 1$, so $i(Q) \geq 3R - 1 = 3i + 2$. The same holds at height -1. Otherwise all vertices have heights in $\{-1/2, 0, 1/2\}$. Only the middle fine row is interior and $i(Q) = 2i + 1$. The two triangles over the middle base give $A \geq 4R$. If ℓ_-, ℓ_+ are the original end-section lengths, concavity gives $\ell_- + \ell_+ \leq 2R$, and $\operatorname{area}(P) = (\ell_- + 2R + \ell_+)/4 \leq R$. Pick now gives $B = A - 4i$, proving the line alternative. □

Outside that line, (10) gives $B \le A - 6i - 2b + 6$; with $B \ge 2b$, this also gives the lower area bound in (3).

3.2 Minimizing sections and the boundary cap

Minimum lattice width is the minimum of $\max_P\langle u, x\rangle - \min_P\langle u, x\rangle$ over nonzero primitive integer covectors u. It is attained: an inscribed ball of radius $r > 0$ gives width at least $2r|u|$, so only finitely many integer directions can improve a fixed direction. An integer unimodular map puts a minimizer on a coordinate axis.

Bohnert's section estimate [5, Lemma 2.12] implies, in that coordinate, that a section of a convex body of width w has length at least $\min\{w/2, \text{distance from the nearer projection end}\}$. This follows from either of his two bounds relative to the longest section. It is preserved by doubling.

Index the fine columns of Q by $0, \dots, W$. An E end has even original fine horizontal coordinate; an O end has odd coordinate. This is only an indexing convention. Original points still have even fine vertical coordinate, and lie on the parity of columns prescribed by E/O. Write u, v for the upper/lower ordinates, $\ell = u - v$. For one concave chain $f = u$ or $f = -v$, let β_j indicate an integer ordinate and c_j an even integer ordinate on a coarse column. On an interior coarse column,

$$i_j = \lceil u_j/2\rceil + \lceil -v_j/2\rceil - 1, \qquad \ell_j \le 2i_j + 2.$$

Lemma 3.2. *For $i \ge 13$, a signature outside the line family satisfies $B - b \le 3i + 6$.*

Proof. Write $N = B - b$. The two narrow nonthin widths are $W = 3, 4$. For $W = 3$, reflect to pattern EO, with coarse interior column 2. Put $M = \lceil f_2/2\rceil$. Concavity and the integer end ordinates give

$$\lceil f_0/2\rceil + f_3 + \beta_1 + \beta_2 - c_2 \le 3M + 1.$$

Indeed $(f_0 + 2f_3)/3 \le f_2$ gives the first two terms at most $3M$. The indicator correction is at most 1 unless f_1, f_2 are integers and $f_2 = 2M - 1$. In that case, writing $f_0 = 2a + p$, $p = 0, 1$, concavity gives $f_1 \ge a + M$ and $f_3 \le 3M - a - 2$, improving the first bound to $3M - 1$. Adding both chains and the extra point at the O end gives $N \le 3i + 6$.

For $W = 4$, pattern EE gives

$$\lceil f_0/2\rceil + \lceil f_4/2\rceil + \beta_1 + \beta_2 + \beta_3 - c_2 \le 2\lceil f_2/2\rceil + 3.$$

The end terms are at most $f_2 + 1$. The indicator sum is at most 2, unless all three are 1 and f_2 is odd; then $f_2 + 1 = 2\lceil f_2/2\rceil$. Adding the chains gives $N \le 2i + 8 \le 3i + 6$. For OO, concavity gives $f_0 + f_4 \le f_1 + f_3$, and $\beta_j - c_j \le 2\lceil f_j/2\rceil - f_j$ on columns 1,3. With $\beta_2 \le 1$, adding the chains and the two O end points gives the same $N \le 2i + 8$.

For the wider cases, let n be the number of interior coarse columns. Their lengths sum to at most $2i + 2n$. End-chord concavity and at most two fine boundary points per interior fine column give

pattern	W	upper bound for N
EE	$2n + 2$	$2i/n + 4n + 6$
OO	$2n$	$4i/n + 4n + 4$
EO	$2n + 1$	$(4n + 2)i/[n(n + 1)] + 4n + 6 - 2/(n + 1)$.

For EE/OO the coarse columns are symmetric, and their average length dominates the average end length. An E end has at most $\ell/2 + 1$ nonoriginal points, an O end at most $\ell + 1$. For EO, the coarse columns $2, 4, \dots, 2n$ instead give

$$\sum \ell_j \ge \frac{n^2}{2n + 1}\ell_0 + \frac{n(n + 1)}{2n + 1}\ell_W.$$

Multiplying by $(2n+1)/(n(n+1))$ bounds $\ell_0/2 + \ell_W$, since $n/(n+1) \geq 1/2$.

For $2 \leq n \leq 8$, all three bounds are at most $3i+6$ when $i \geq 13$. For $n \geq 9$, the minimizing section bound, in original units, gives

$$i \geq \sum_{j=1}^{n} (\min\{j, n+1-j\} - 1) = \lfloor (n-1)^2/4 \rfloor \geq (n^2 - 2n)/4.$$

Substitution reduces the three comparisons to $3n^2 - 24n + 4 \geq 0$, $3n^2 - 26n + 16 \geq 0$, and $3n^3 - 23n^2 - 16n + 12 \geq 0$. Each is positive at 9 and increasing thereafter. Widths below 3 have no coarse interior column or are the thin line case. □

3.3 A joint bound at larger widths

Lemma 3.3. *If $i \geq 17$ and the minimum lattice width of P exceeds 2, then $L_P(2) \leq 6i + 2b + 6$, equivalently $A + B \leq 12i + 4b + 10$.*

Proof. Use the minimizing fine projection and E/O convention just introduced. Its patterns and coarse columns are

	W	interior coarse columns	coarse ends
EE	$2n+2$	$2, 4, \ldots, 2n$	$0, W$
EO	$2n+1$	$2, 4, \ldots, 2n$	0
OO	$2n$	$1, 3, \ldots, 2n-1$	none.

Here $n \geq 2$, and in the wide OO case $n \geq 3$. Let p_j count all closed fine points on column j, and let i_j, b_j be its original interior and boundary counts on coarse columns. Then $\sum p_j = L_P(2)$, $\sum i_j = i$, $\sum b_j = b$. All variables are nonnegative. Convexity and interval counting give

$$\ell_{j-1} - 2\ell_j + \ell_{j+1} \leq 0, \qquad \ell_j \geq \min\{W/2, j, W-j\}, \tag{11}$$

$$p_j \leq \ell_j + 1 \quad \text{on all columns,}$$

$$\ell_j \leq 2i_j + 2, \quad 2i_j \leq \ell_j + 2 \quad \text{on interior coarse columns,}$$

$$p_j \leq 2i_j + b_j + 1, \quad b_j \leq 2 \quad \text{on interior coarse columns,} \tag{12}$$

$$\ell_j \leq 2b_j, \quad p_j \leq 2b_j + 1 \quad \text{on coarse ends.} \tag{13}$$

For the closed coarse bound, between two odd fine points is an even point strictly inside the section, so there are at most $i_j + 1$ odd points. At an end, the interval has integer endpoints; its even points give the two end bounds even when it is a single odd vertex.

For $2 \leq n \leq 8$, relax these conditions to nonnegative real variables and add $\sum i_j \geq 17$. Write the displayed constraints as $Mx \leq r$, with objective $cx = \sum p_j - 6\sum i_j - 2\sum b_j$. The companion file `dual_weights.json` contains rational vectors $\lambda \geq 0$ with

$$M^{\mathsf{T}}\lambda \geq c^{\mathsf{T}}, \qquad r^{\mathsf{T}}\lambda = \gamma \leq 6.$$

Their exact values are checked by the supplied rational verifier. The bounds are

n	EE	EO	OO
2	-4	-5	-6
3	$-17/3$	$-20/3$	$-23/3$
4	$-9/2$	$-11/2$	$-13/2$
5	$-11/5$	$-16/5$	$-21/5$
6	$2/3$	$-1/3$	$-4/3$
7	$7/3$	$8/3$	$5/3$
8	$9/2$	$94/21$	$7/2$

For every feasible $x \geq 0$, $cx \leq \lambda^{\mathsf{T}} M x \leq \lambda^{\mathsf{T}} r$. Thus these certificates prove the bound for all $i \geq 17$ at the indicated indices. No floating optimization result is required. The extra OO $n = 2$ certificate is not needed for the wide case.

For $n \geq 9$, let I_j count strict fine points on a coarse interior column; interval parity gives $I_j \leq 2i_j + 1$. Take the hull of the integer endpoints on columns $j-1, j+1$. For a two-dimensional integer trapezoid of width 2, Pick gives

$$p_{j-1} + p_{j+1} = p_j(T) + I_j(T) + 2 \leq p_j + I_j + 2.$$

This is the row identity of [7, Lemma 6.5]. If the hull is a segment, its two end sections are single points, so the inequality holds directly. The minimum-section bound ensures all adjacent sections contain fine points.

Summing these inequalities and subtracting the twice-counted intervening noncoarse columns gives

$$L_P(2) - 6i - 2b \leq \begin{cases} 5n + 2 - S_{\mathrm{EE}}, & \mathrm{EE}, \\ 5n + 1 - S_{\mathrm{EO}}, & \mathrm{EO}, \\ 5n - S_{\mathrm{OO}}, & \mathrm{OO}. \end{cases}$$

Here the first two S's sum p_j over $j = 3, 5, \ldots, 2n-1$; the third sums over $j = 2, 4, \ldots, 2n-2$. Indeed each coarse interior contributes $2p_j + I_j + 2 \leq 6i_j + 2b_j + 5$, and each coarse end contributes one additional constant. Since $p_j \geq \min\{j, W - j\}$, pairing columns gives

$$S_{\mathrm{EE}} \geq \frac{n^2 + 2n - 3}{2}, \quad S_{\mathrm{EO}} \geq \frac{n^2 + n - 2}{2}, \quad S_{\mathrm{OO}} \geq \frac{n^2 - 1}{2}.$$

The exact EE sum is $n^2/2 + n - 1$ for even n, the stated value for odd n; the EO sum is exact and the OO sum is $\lfloor n^2/2 \rfloor$. The resulting excess bounds are

$$\frac{-n^2 + 8n + 7}{2}, \quad \frac{-n^2 + 9n + 4}{2}, \quad \frac{-n^2 + 10n + 1}{2}.$$

At $n = 9$ these are $-1, 2, 5$, and their successive differences are $(7 - 2n)/2, (8 - 2n)/2, (9 - 2n)/2$, all negative thereafter. This closes the tail and the proof. □

4 Narrow widths and the exceptional area correction

Width below 1 has no original interior line. At width 1 the admissible case is the line family. At width 3/2, write the doubled projection as $[-2, 1]$. There is one coarse interior row 0. The end and middle bounds

$$p_{-2} \leq 2b_{-2} + 1, \quad p_0 \leq 2i + b_0 + 1, \quad I_0 \leq 2i + 1, \quad p_{-1} + p_1 \leq p_0 + I_0 + 2$$

give $L_P(2) \leq 6i + 2b + 6$. Width 2 can exceed this by one.

4.1 The width-two equality family

Lemma 4.1. *At minimum width 2, the EE pattern satisfies $L_P(2) \leq 6i + 2b + 7$. Its equality signatures are exactly (6), and all are realized for every $i \geq 2$. The OO pattern satisfies*

$$L_P(2) - 6i - 2b \leq 12 - i - b.$$

For $i \geq 17$, all excess-seven signatures therefore belong to (6).

Proof. In the EE pattern index the projection of Q by $0, \dots, 4$. All original interior points are on column 2. Interval parity and the row hull inequality give

$$L_P(2) \le (2b_0 + 1) + (2b_4 + 1) + 2(2i + b_2 + 1) + (2i + 1) + 2 = 6i + 2b + 7.$$

Equality forces both ordinates at each end to be odd integers. Translate vertically so the middle original interior points have fine ordinates $2, 4, \dots, 2i$. Equality in the strict fine count then forces

$$0 \le v_2 < 1, \qquad 2i + 1 < u_2 \le 2i + 2.$$

Put $c_L = 1$ if $v_2 = 0$, otherwise 0, and $c_U = 1$ if $u_2 = 2i + 2$, otherwise 0. Then $b_2 = c_L + c_U$. Writing $L_j = \lceil v_j \rceil, U_j = \lfloor u_j \rfloor$ for $j = 1, 3$, equality in the row hull bound is equivalent to

$$L_1 + L_3 = 1 - c_L, \qquad U_1 + U_3 = 4i + 3 + c_U. \tag{14}$$

The hull has half-integer middle endpoints, and equality forces its strict and closed middle counts to equal those of Q; these are the only possible endpoint sums. Conversely these conditions give $p_1 + p_3 = 4i + b_2 + 4$, hence equality.

Let β_U, β_L count integer upper/lower ordinates on columns 1,3, and put $\beta = \beta_U + \beta_L$. If $c_U = 1$, concavity and (14) force both upper ordinates to be integer. If $c_U = 0$ and neither were integer, the upper chain would have no vertex on columns 1,2,3, so it would join its two odd end ordinates by one edge, making u_2 integral, a contradiction. Thus $\beta_U \ge 1 + c_U$, and likewise $\beta_L \ge 1 + c_L$. Consequently

$$d = \beta - b_2 - 2 \in \{0, 1, 2\}, \qquad B = 2b + 2 + \beta - b_2 = 2b + 4 + d.$$

Pick with the equality value of $L_P(2)$ gives $A = 12i + 2b + 8 - d$.

For the boundary range, let $g_U = 1$ if $\beta_U = 2$, otherwise 0, and define g_L similarly. Concavity and odd end rounding give

$$u_0 + u_4 \le 4i + 2 + 2c_U - 2g_U, \qquad v_0 + v_4 \ge 2 - 2c_L + 2g_L.$$

To see the improvement when $g_U = 1$, concavity gives $u_0 \le (3U_1 - U_3)/2$, $u_4 \le (3U_3 - U_1)/2$. Rounding down to odd integers loses 2 in total if $U_1 + U_3 = 4i + 4$, and 3 if the sum is $4i + 3$. The lower-chain rounding is the reverse. It follows that $b \le 2i + 2b_2 - g_U - g_L$. Since $g_U + g_L \ge \beta - 2$,

$$b + 2d \le 2i + 2\beta - g_U - g_L - 4 \le 2i + \beta - 2 \le 2i + 2.$$

For attainment, choose parameters as follows:

d	c_L	c_U	left capacity	right capacity
0	1	1	i	i
1	1	0	$i - 1$	i
2	0	0	$i - 1$	$i - 1$.

Set $s = 2 - d$, $b_0 = \min\{\text{left capacity}, b - s\}$, $b_4 = b - s - b_0$. The range in (6) ensures the capacities. Take the lower chain

$$(0, 1), (1, 0), (3, 1 - c_L), (4, v_4)$$

and upper chain

$$(0, 1 + 2b_0), (1, U_1), (3, 2i + 2), (4, v_4 + 2b_4),$$

where $v_4 = 1$ for $d \le 1$, $v_4 = 3$ for $d = 2$, and $U_1 = 2i + 2$ for $d = 0$, $U_1 = 2i + 1$ otherwise. Their slopes have the required monotonicity by the capacity bounds. Repeated endpoints are omitted. The original counts are i, b, the conditions (14) hold, and $\beta = 4$, giving the desired A, B. The inner rectangle for $d = 0, 1$ has height at least $2i + 1$. For $d = 2$, the inner parallelogram has width $|2a + c| + (2i + 1)|c|$ in direction (a, c). Any direction with $c \neq 0$ therefore has fine width at least 5, whereas the horizontal direction has width 4. Thus the minimum original width is 2.

For OO, the original columns are 1,3. Row pairing gives

$$L_P(2) \le 2p_1 + I_1 + 2p_3 + I_3 + 4 - p_2 \le 6i + 2b + 10 - p_2.$$

The coarse points in a section span length at least $2(i_j + b_j - 1)$, including the empty case as a vacuous lower bound. Concavity gives $\ell_2 \ge i + b - 2$, whence $p_2 \ge i + b - 2$. This proves the OO bound. For smaller i, its only pairs with possible excess above 6 are $(2, 2), (2, 3), (3, 2)$, included in the complete finite source comparison of Section 6. □

4.2 The correction at $b = 2, B = 4$

Lemma 4.2. *If $i \ge 17$ and a minimizing original projection has at least two interior integral rows, then $A < 12i + 2b + 6$.*

Proof. Let these rows be $1, \dots, n$, $n \ge 2$, and put $\ell_y = \text{length}(P \cap \{y = \text{constant}\})$. The half-row bands centered at $j = 1, \dots, n$ have normalized area at most $8\ell_j \le 8(i_j + 1)$. The two exterior half-bands have total area at most $2(b_0 + b_{n+1} + \ell_{1/2} + \ell_{n+1/2})$. Concavity gives $\ell_{1/2} + \ell_{n+1/2} \le \ell_k + \ell_{n-k+1}$. Among the $\lfloor n/2 \rfloor$ disjoint pairs, one has interior count at most $i/\lfloor n/2 \rfloor$. Hence

$$A \le 8i + 8n + 2b + \frac{2i}{\lfloor n/2 \rfloor} + 4. \tag{15}$$

For $n = 2$ this is strictly below $12i + 2b + 6$ when $i \ge 8$. For $n \ge 3$, it suffices that

$$i > f(n) = \frac{4n^2 - 5n + 1}{2n - 4}.$$

At $n = 3, \dots, 7$, the values are $11, 45/4, 38/3, 115/8, 81/5$, all below 17. At $n = 8, 9, 10$ they are $217/12, 20, 351/16$. The only remaining $i \ge 17$ are therefore among 17,...,21. The complete maximal-polygon classification gives the global minimum-width maxima $7, 15/2, 8, 15/2, 8$ at those five i, respectively [3, proof of Theorem 1.4, pp. 9–10]. They exclude the required $n = 8, 9, 10$. Their catalogue scope is explained in Section 6.

For $n \ge 11$, the earlier section sum gives $i \ge (n^2 - 2n)/4 > f(n)$: the latter strict comparison is equivalent to $n^3 - 12n^2 + 14n - 2 > 0$, which is 31 at 11 and has increment $3n^2 - 21n + 3 > 0$ thereafter. □

Lemma 4.3. *If $i \ge 17, b = 2, B = 4$, then $A \le 12i + 8$.*

Proof. With at least two minimizing interior original rows, Lemma 4.2 and even A give the result. Width 1 is the line case and gives $A = 4i + 4$. The remaining one-row patterns are EE of fine height 4 and EO of fine height 3. Integral polygons cannot have $b = 2$.

For EE put the fine end rows at $-2, 2$, and let $c = 0, 1, 2$ be the original boundary count on row 0. If H_-, H_+ are the fine end lengths, the budget of four fine boundary points gives $H_- + H_+ \le 2 - c$. In original units, trapezoid integration and concavity give

$$A \le 12\ell_0 + 2(\ell_{-1} + \ell_1).$$

The original middle endpoints have fractional parts in $\{0, 1/6, 1/4, 1/3, 1/2, 2/3, 3/4, 5/6\}$, by interpolating between fine vertex rows $-2, -1, 1, 2$. Thus $\ell_0 \le i + 2/3 + c/6$. Equality in the area concavity step makes both chains affine across the fine rows $-1, 0, 1$.

If $c = 0, H_- + H_+ \le 1$, the bound is $12i + 9$, and evenness gives $12i + 8$. If the sum is 2, all four fine points lie on the ends; there is no interior-row vertex and $A = 8$. If $c = 2$, both ends are points and the two other fine boundary points are the original middle endpoints. No odd-row vertex remains, so $A = 4(2i + 2) = 8i + 8$. If $c = 1, H_- + H_+ = 1$, again no odd-row vertex remains. The opposite middle endpoint is a nonintegral fine half-integer; the middle fine length is at most $2i + 3/2$, giving $A \le 8i + 8$.

For $c = 1, H_- + H_+ = 0$, an additional middle fine boundary point must be odd, giving $A \le 12i + 6$. Otherwise exactly one fine boundary point is on an odd row, and the general bound is $12i + 10$. Equality would make the chain through the original middle point affine: if it is integer on one odd row, it is integer on the other, requiring two fine boundary points there. If neither is integer, it has no vertex there or at the middle under this affineness condition. Its end points have opposite fine horizontal parity, so their middle average is nonintegral, contradicting the original middle point. Equality is impossible, and evenness again gives $A \le 12i + 8$.

For EO put the fine ends at $-2, 1$, let c be the middle original boundary count and H the lower fine end length. The lower original count is $2 - c$, and $H \le 2 - c$. If $c = 0$, then $H = 2$, all fine boundary is at the ends, and $A = 6$. If $c = 2$, $H = 0$, the lower end is odd, the upper is a point, and no vertex remains at row -1; thus $A = 3(2i + 2)$. For $c = 1, H = 1$, the same absence of an odd-row vertex gives $A = 2 + 3H_0 \le 6i + 7$, where $H_0 \le 2i + 5/3$ is the fine middle length. For $c = 1, H = 0$, a two-point upper end gives $A = 3H_0 + 1 \le 6i + 6$. If instead the upper end is a point and both middle endpoints are fine, the nonoriginal one is odd, giving $A = 3H_0 \le 6i + 3$.

The last EO case has four fine points: an original lower point E, an original middle point V, an upper point T and a point W on row -1. An integer shear and translation put $E = (0, -2), V = (0, 0)$. Reflect horizontally so $T = (a, 1)$, $a > 0$. Convexity gives $W = (c', -1)$, $-a \le c' \le 0$. The middle fine interval is $[0, 2a/3]$. The edge ET is primitive, so $3 \nmid a$ and $i = \lceil a/3 \rceil - 1$, whence $a = 3i + 1$ or $3i + 2$. Shoelace gives $A = 2(a - c') \le 4a \le 12i + 8$, finishing every case. □

For $i \ge 17$, the marginal bounds, Lemma 3.1, Lemma 3.2, Lemma 3.3 and the narrow-width analysis put every signature in the body, top or line. The generic maximum-area correction is (9). At $b = 2$, Lemma 4.3 excludes $B = 4$ at the two even areas $12i + 10, 12i + 12$, giving precisely the two corrections in (5). This proves necessity in Theorem 2.1 for $i \ge 17$.

5 Uniform witnesses

Every polygon in this section is $P = Q/2$, where Q is the hull of the listed integer points. Collinear or repeated outline points are removed. For successive collinear primitive-direction subedges of lengths p, q, their boundary gcds add to $p+q$; hence merging them does not alter the boundary computations. A listed original boundary point need not remain an extreme vertex. These conventions include all parameter endpoints.

5.1 The line family

For $4i + 4 \le A \le 8i + 8$, set $n = A - 4i - 4$, $r = \lfloor n/2 \rfloor, s = \lceil n/2 \rceil$, and take

$$Q = \operatorname{conv}\{(0, -1), (s, -1), (2i + 2, 0), (r, 1), (0, 1)\}.$$

Since $0 \le r, s \le 2i+2$, this is convex and two-dimensional. The original middle section is $[0, i+1]$; its endpoints are the two boundary points, with i interior points. Integration gives $A = r+s+4i+4$, and $i(Q) = 2i+1$ gives $B = A-4i$ by Pick. This realizes the entire line family. Its members are distinct by area.

5.2 Body coordinates and the two outer layers

For a body triple put

$$m = 3i - b + 6, \quad e = A \bmod 2, \quad j = \frac{A - 6i - 4b + 6 - e}{2}, \quad k = \frac{B - 2b - e}{2}.$$

Its conditions become

$$j, k \ge 0, \quad k \le j, \quad 2k + e \le m, \quad j + k + e \le m + 2. \tag{16}$$

For $m \ge 2$, the area bound gives $j \le m+1$ at $e = 0$ and $j \le m$ at $e = 1$. It excludes $(e, j, k) = (0, m+1, 0)$ except at $b = 2i+4$, and also excludes $(0, m, 0)$ at $b = 2$. At $m = 0$ only $j = k = e = 0$ occurs; at $m = 1$, $k = 0$ and $j, e \in \{0, 1\}$. We first realize the layers $j + k + e = m + 2, m + 1$.

If $j + k + e = m + 2$, then $k \ge 1$. Put $f = 2k + e - 2$ and take

$$Q = \mathrm{conv}\{(0,1), (f,1), (4i+4-f, -1), (2b-3, -2), (1, -2), (0, -1)\}. \tag{17}$$

Here $0 \le f \le m-2$. Right-chain convexity requires $2b - 3 \le 6i + 6 - 2f$, which follows strictly from this bound. The left chain is convex as well. The original bottom has $b-2$ boundary points; the middle fine section is $[0, 2i+2]$, giving i interior and two original boundary points. On fine rows $-2, -1, 0, 1$, the closed counts are

$$2b - 3, \quad 4i + 5 - f, \quad 2i + 3, \quad f + 1.$$

Thus $L_P(2) = 6i + 2b + 6$, $B = 2b + f + 2 = 2b + 2k + e$, and

$$A = 2L_P(2) - B - 2 = 12i + 2b + 10 - 2k - e = 6i + 4b - 6 + 2j + e.$$

On $j + k + e = m + 1$, first suppose $b \ge 3$, $2k + e \ge 1$. With $f = 2k + e - 1$, take

$$Q = \mathrm{conv}\{(0,1), (f,1), (4i+4-f, -1), (2b-5, -2), (0, -2)\}. \tag{18}$$

Its mixed-parity bottom has $b-2$ original boundary points, the middle has i interior and two boundary points, and right convexity follows from $f \le m-1$. The four fine row counts are $2b-4, 4i+5-f, 2i+3, f+1$. Hence $L_P(2) = 6i + 2b + 5$, $B = 2b + f + 1$, and Pick gives the target A.

For $b = 2, k \ge 1, 2k + e \le m - 1$, put $f = 2k + e - 2$ and take

$$Q = \mathrm{conv}\{(0,1), (f,1), (4i+3-f, -1), (1, -2), (0, -2)\}.$$

The bottom contains one original boundary point and the middle fine section is $[0, (4i+3)/2]$, with i original interior points and one original boundary point. The condition $f \le 3i + 1$ gives convexity. The fine counts $2, 4i + 4 - f, 2i + 2, f + 1$ give $L_P(2) = 6i + 9$, $B = f + 6 = 4 + 2k + e$, and the required area.

At $b = 2, 2k + e = m$, necessarily $j = k + 1$; use

$$Q = \mathrm{conv}\{(0, 1), (3i, 1), (3, -2), (1, -2), (0, -1)\}. \tag{19}$$

Its middle fine section is $[0, 2i + 1]$, and the original bottom point is $(2, -2)$ in fine coordinates. Thus the counts are $i, b = 2$. The edge gcds are $3i, 3, 2, 1, 2$, so $B = 3i + 8$; shoelace gives $A = 9i + 8$, exactly the desired signature.

For $e = k = 0$, the first outer layer is allowed only at $b = 2i + 4$. The triangle

$$Q = \mathrm{conv}\{(-1, -2), (4i + 3, -2), (1, 2)\}$$

has i original interior points, $b = 2i + 4$, $B = 4i + 8$, $A = 16i + 16$. It supplies that exception. The remaining first-layer case $b = 2, e = 1, k = 0$ is covered by the high-difference construction below.

5.3 The bulk

It remains to cover $j + k + e \le m$. For $b \ge 3$, if $e = 0$ or $e = 1, j \ge k + 1$, take

$$Q = \mathrm{conv}\{(0, 1), (2k, 1), (2i + 2, 0), (i + b - 2 + j - k, -1), (2b - 6 + e, -2), (0, -2)\}.$$

The right-middle cross products are $j + k - m$ and $e - 2(j - k)$, both nonpositive; the other corners have the required sign. The original middle section is $[0, i + 1]$, and the bottom contains $b - 2$ original boundary points. Edge gcds give $B = 2b + 2k + e$; shoelace gives $A = 6i + 4b - 6 + 2j + e$. For $e = 1, j = k$, instead take

$$Q = \mathrm{conv}\{(0, 1), (2k + 1, 1), (2i + 2, 0), (2b - 6, -2), (0, -2)\}.$$

Its right convexity condition is $2k + 1 \le m$, and the original counts are unchanged. Here $B = 2b + 2k + 1$, $A = 6i + 4b - 5 + 2k$. These formulas include the allowed $m = 0, 1$ cases.

For $b = 2, j - k \ge 2, 2k + e \ge 1$, use

$$Q = \mathrm{conv}\{(0, 1), (2k + e - 1, 1), (2i + 2, 0), (i + j - k, -1), (1, -2), (0, 0)\}. \tag{20}$$

It is convex whenever $j + k + e \le m + 1$ and $3 - 2(j - k) \le 0$, covering both its bulk uses and the remaining first-layer case. The fine middle endpoints are $0, 2i + 2$; the odd bottom vertex has no original point. Thus the original counts are $i, 2$. Gcds and shoelace give $B = 4 + 2k + e$, $A = 6i + 2 + 2j + e$.

For $b = 2, e = k = 0$, the exclusions give $0 \le j \le 3i + 3$. Put $h = 3i + 1, H = 0$ at $j = 0$, and $h = 3i + 2, H = j - 1$ at $j \ge 1$. Take

$$Q = \mathrm{conv}\{(0, 0), (1, H), (2, 0), (3, -h)\}. \tag{21}$$

The condition $H \le h$ gives convexity, including the collinear endpoint. The upper original triangle lies between $x = 0, 1$, with apex at $x = 1/2$, so it has no original interior point. On original row $y = -r$, the lower triangle has endpoints $3r/h, 1 + r/h$. Its only possible original interior coordinate is $x = 1$, which is inside exactly for $0 < r < h/3$. Hence its count is $\lceil h/3 \rceil - 1 = i$. The two original boundary points are the base endpoints. Every nonbase outline subedge is primitive, since $\gcd(3, h) = 1$, giving $B = 4$; when $H = 0$, the base has fine length 2 and gives the same count. The area is $A = 2(h + H) = 6i + 2 + 2j$. At $H = h$, the fine point $(2, 0)$, corresponding to an original lattice point, lies on a merged edge rather than remaining a vertex. Its boundary status is unchanged. In particular, this body family includes $A = 8i, B = 4$, satisfying (2).

Finally let $b = 2, \delta = j - k \in \{0, 1\}$, $t = 2k + e \le 3i + 4$. If $t \le 3i + 1 + \delta$, put $D = 6i + 2 + 2\delta - 2t$ and take

$$Q = \operatorname{conv}\{(0, 0), (0, 1), (t, 1), (D, -2)\}.$$

Here D is even and nonnegative, giving convexity. Its original middle interval is $[0, i + (1 + \delta)/3]$, so it has i interior points and one boundary point there, and the bottom supplies the other boundary point. The right edge is primitive because $D - t$ is 2 or 1 modulo 3. Thus $B = t + 4$, and $A = 3t + D = 6i + 2 + t + 2\delta$.

For $\delta = 0, t = 3i + 4$, use

$$Q = \operatorname{conv}\{(0, 1), (3i - 2, 1), (7, -2), (3, -2)\}.$$

Its fine middle interval is $[1, 2i + 1]$; the bottom odd-ended interval has two original points. The edge gcds give $B = 3i + 8$, and $A = 9i + 6$. For $t = 3i + 3$, replace the upper-left corner by $(1, 1), (1, 0)$. For $t = 3i + 2$, also replace the upper-right corner by $(3i - 3, 1), (2i + 1, 0)$. Each removed primitive triangle has normalized area 1; replacing its two primitive old edges by one reduces B by 1 as well. All removed and new fine points are nonoriginal, so i, b do not change. The top segment stays nonempty at $i = 2$.

For $\delta = 1, t = 3i + 4$, use (19). For $t = 3i + 3$, shave its upper-right primitive corner, replacing $(3i, 1)$ by $(3i - 1, 1), (2i + 1, 0)$. Again A, B both decrease by 1 without changing i, b. Together these families exhaust (16). Thus every body signature is realized for every $i \ge 2$. The top witnesses were already constructed in Lemma 4.1.

6 Computational inputs and reproduction

The complete finite input is the published classification [4, Classification 6.7 and the following enumeration], with the released dataset [8]. Its convention is k-rational, meaning kP is integral, so $k = 2$ includes integral polygons. Affine equivalence uses integer translations and unimodular matrices; these preserve $L_P(t)$ for every integer t.

We explain the completeness and count parameters used here. Every positive-i half-integral polygon is contained in a same-i 2-maximal polygon. Indeed the minimizing section estimate gives, for every positive i,

$$i \ge \lfloor (n - 1)^2/4 \rfloor \ge (n^2 - 2n)/4$$

when there are n interior integral columns. Since $w \le n + 1$, this implies $w \le 2 + \sqrt{4i + 1}$. Some integral section has length at most $i + 1$. A chord to a longest section, at distance at least $1/2$ from the projection end, gives $\ell_{\max} \le 2w(i + 1)$. Thus $\operatorname{area}(P) \le 2w^2(i + 1)$, a bound depending only on i. The integer normalized area increases under proper half-integral containment, so a same-i enlargement chain terminates. Successively deleting a fine vertex outside a target subpolygon reaches every subpolygon; along a same-i chain no interior count is lost. This is the shaving principle in [4, Proposition 2.9, Algorithm 1 and Remark 2.14].

The maximal catalogue combines strip polygons with Fine interiors. For an integral polytope K, its Fine interior is

$$F(K) = \{x : \langle x, u \rangle \ge \min_{y \in K} \langle y, u \rangle + 1 \text{ for every } u \in \mathbb{Z}^{\dim K} \setminus \{0\}\}.$$

Its integer points are exactly the interior integer points of K. For a half-integral polygon P_0, the lattice pyramid with base $2P_0$ at height -1 and apex $(0, 0, 1)$ has middle section P_0. Its interior integer points lie at height 0 and number $i(P_0)$. Positive interior count forces every lattice width

to be at least 2, and the height direction has width 2. Its Fine interior lies in the middle plane. Consequently its Fine polygon has exactly $i(P_0)$ total original lattice points, including its own boundary.

The source results [4, Propositions 6.4 and 6.5] and [6, Theorems 2.5, 3.2 and 4.1; Remark 3.16] put the one-dimensional Fine case in the strip branch and reconstruct the two-dimensional case from the complete Fine catalogue through 40 total points. Hence the maximal catalogue through 40 interior points covers all arrangements, not only collinear ones. The same-i containment argument and monotonicity of minimum width justify the five width maxima used in Lemma 4.2.

For $i = 2, \dots, 16$, version 1.0.0 of the dataset provides the full function files `ehrhart_iN.h5`; its separate collinear polygon files are not used. Each normalized coefficient matrix is stored as an integer 3×2 array

$$\begin{pmatrix} A & A \\ 4b & 2B \\ 8i + 4b - A & 8 \end{pmatrix}.$$

The decoder verifies all entries, the area group, parity and uniqueness, then filters only $b \geq 2$. The resulting exact sets S_i have empty symmetric difference with C_i at every one of these fifteen i. For reference, their cardinalities are

i	2	3	4	5	6	7	8	9
$\lvert S_i \rvert$	408	761	1267	1953	2846	3973	5361	7037

i	10	11	12	13	14	15	16
$\lvert S_i \rvert$	9028	11361	14063	17161	20682	24653	29101.

The certificate is equality of the full sets, not just these counts. Combined with the published completeness assertion this proves finite necessity. With the analytic necessity for $i \geq 17$ and the constructions in Section 5, it completes Theorem 2.1.

The companion Python source includes the exact section model and its 21 rational dual weights, a verifier using integer and rational arithmetic, the source-file checksum manifest and coefficient-set comparator, and the line/body/top constructors. Polygon counts use integer half-plane intersection, shoelace area and edge gcds; small cases are also checked by independent bounding-box enumeration. Data retrieval is pinned to version 1.0.0 and its official checksums. Exact source-set and inequality verification can be reproduced without rerunning the polygon classification. The companion archive `half-integral-ehrhart-verification.zip` includes a root `README.md` with fixed-input retrieval, full-comparison commands and pinned optional HDF5 dependencies. From the archive root, run `python3 numerical_case/verify.py --output dual.json` for the certificate-only check.

The published catalogue computations are cited external inputs and have not been independently reconstructed here. Floating linear optimization was used only to discover rational weights; the retained rational inequalities suffice for their verification. Finite polygon tests check the implementation, whereas the parameter arguments in the preceding sections establish universal attainment.

7 The exact count

For a body triple recall $m = 3i - b + 6$ and the coordinates (16). As b runs from 2 to $3i + 6$, m runs from $N = 3i + 4$ to 0. Temporarily remove the generic maximum-area point $(e, j, k) = (0, m + 1, 0)$. For $m \geq 2$, at $e = 0, k = 0$ there are $m + 1$ choices, and at $e = 0, k \geq 1$ there are $m + 3 - 2k$, for $1 \leq k \leq \lfloor m/2 \rfloor$. At $e = 1, k = 0$ there are $m + 1$ choices, and at $e = 1, k \geq 1$ there are $m + 2 - 2k$,

for $1 \le k \le \lfloor (m-1)/2 \rfloor$. Summing gives $m(m+7)/2$ for both parities. The $m = 1$ count is 4 and $m = 0$ count is 1. Reinstating the exceptional point at $b = 2i + 4$ adds one, while the additional $b = 2, j = m, k = 0, e = 0$ exclusion removes one. Therefore

$$\#\text{body} = 1 + \sum_{m=1}^{N} \frac{m(m+7)}{2} = 1 + \frac{N(N+1)(N+11)}{6} = \frac{9}{2}i^3 + 36i^2 + \frac{155}{2}i + 51.$$

The top count is $(2i+1) + (2i-1) + (2i-3) = 6i - 3$, and the line count is $4i + 5$. These sets are disjoint: the body has $A + B \le 12i + 4b + 10$, the top has $A + B = 12i + 4b + 12$, and their respective $A - B$ values are at least $6i + 2b - 6 > 4i$ and $12i + 4 - 2d > 4i$, whereas the line has $A - B = 4i$.

Proof of Theorem 1.1. Theorem 2.1 identifies the attained signatures with C_i. The constituent formula (1) identifies signatures with entire functions at fixed i. Adding the three disjoint counts gives

$$\mathcal{E}(i) = \frac{9}{2}i^3 + 36i^2 + \frac{175}{2}i + 53 = \frac{9i^3 + 72i^2 + 175i + 106}{2}.$$

All parameter ranges include $i = 2$, integral polygons and period collapse. □

AI-use disclosure. OpenAI Codex assisted with proof development, exact verification code and manuscript drafting. The cited computational classifications remain external dependencies. The mathematical arguments and accompanying verification inputs are supplied for independent checking.

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